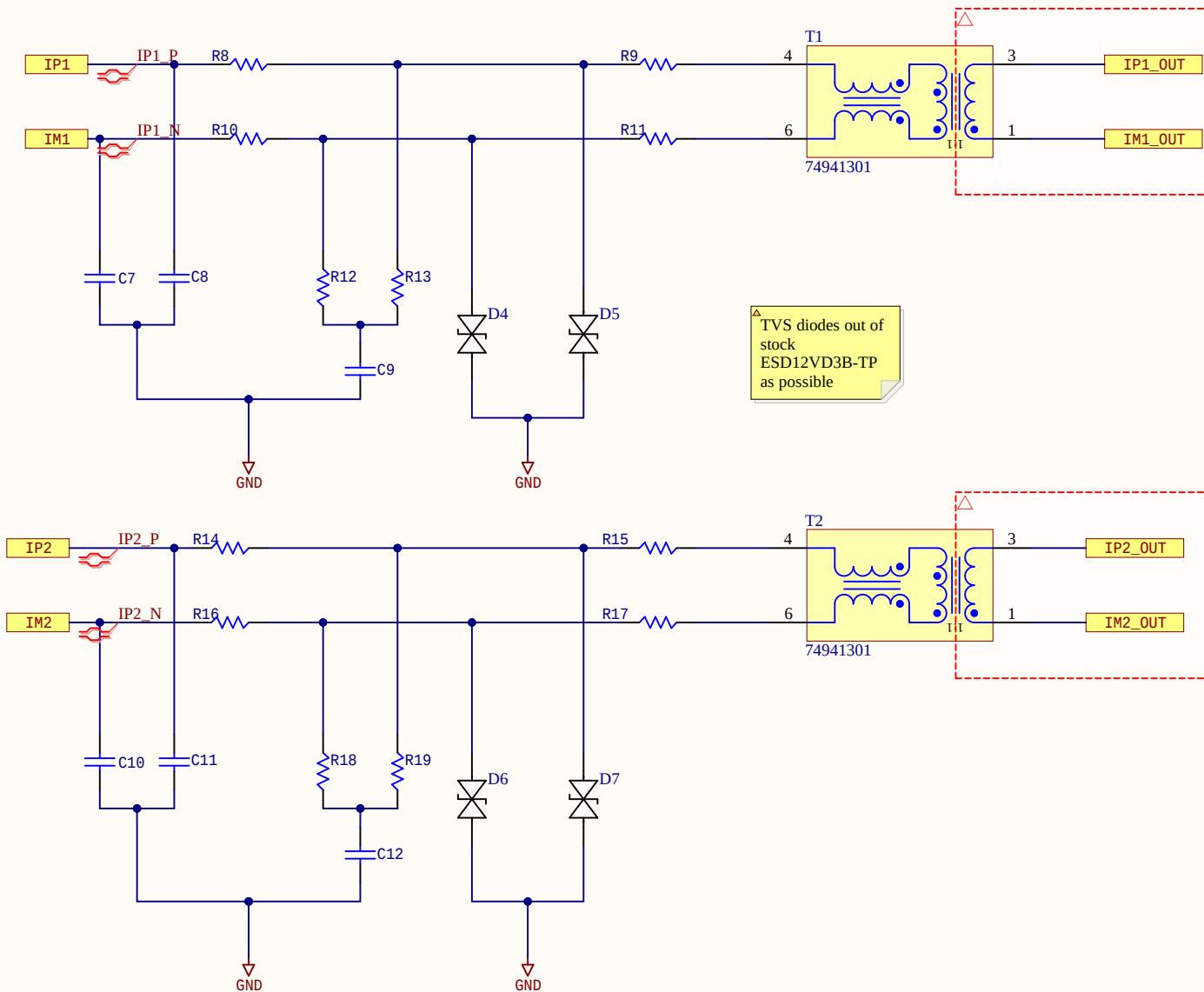


A

B

C



Description
SPI Mode Select. These pins select SPI controller mode (MSTR connected to VDD5 and MSTR2 connected to VDD52) or peripheral mode (MSTR and MSTR2 connected to GND).
SPI Controller Outputs (Controller Mode) or Peripheral Inputs (Peripheral Mode).
SPI Controller Inputs (Controller Mode) or Peripheral Outputs (Peripheral Mode).
SPI Clock Inputs (Controller) or Outputs (Peripheral).
Active Low SPI Chip-Select Inputs (Controller Mode) or Outputs (Peripheral Mode).
SPI Power-Supply Inputs (1.7 V to 5.5 V).
LPCM Interrupt Current-Limited Outputs.
Device Wake-Up State Current-Limited Outputs or Inputs.
Ground. The GND pins must be shorted together external to the IC.
Isolated Interface Minus Inputs/Outputs.
Isolated Interface Plus Inputs/Outputs.
isoSPI Power-Supply Inputs (3.0 V to 5.5 V).
High Voltage Power-Supply Inputs (3.0 V to 30 V).
Multilevel, Resistor Set SPI PHA/POL Selection Inputs.
Multilevel, Resistor Set LPCM Timeout Selection Inputs.
Multilevel, Resistor Set Transceiver Mode Selection Inputs.
Exposed Pad. Must be connected to GND.

B

- A

MSTR -> Master (mode select)
PICO -> Peripheral In, Controller Out
POCI -> Peripheral Out, Controller In
SCK -> Serial Clock
CS -> Chip Select
VDD5 -> SPI power supply
INTR -> Interrupt
WAKE -> Wake pin
GND -> Ground
IM -> IsoSPI Minus
IP -> IsoSPI Plus
PHAPOL -> Phase/Polarity (SPI idle L/H, SPI latch rise/fall)
RTO -> Resistor set Timeout
XCVRMD -> Transceiver Mode
EP -> Exposed

C

A

ADBMS battery monitors use SPI Mode 3" Page 19

Table 20. PHAPOL SPI Modes

SPI Mode	POL	PHA	Description	PHAPOL Pin Connection
0	0	0	SCK idles low, latches on rising (first) edge	GND
1	0	1	SCK idles low, latches on falling (second) edge	20 kΩ resistor to GND

D

2

3

4

5

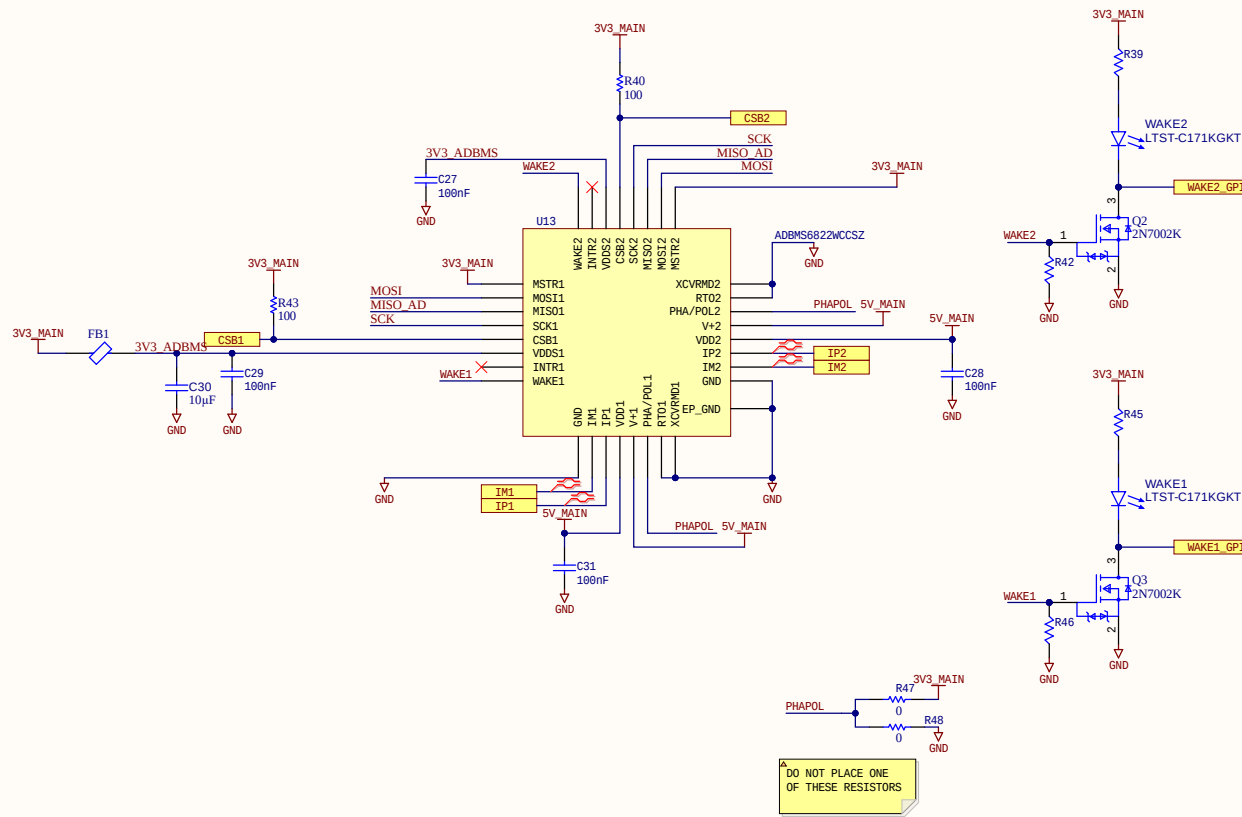


Table 21. XCVRMD Pin Tra

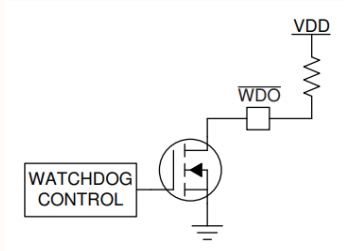
Description
Standard Bidirectional isoSP
Standard Bidirectional isoSP
Monitor Support
4 Mbps Unidirectional
2 Mbps with 1-Bit Latency

Title

A

B

C



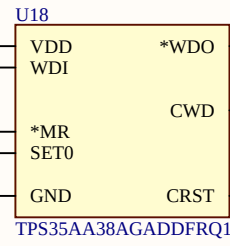
Non-Inverting Open Drain Buffer

BMS_FAULT_GPIO TP20

Might have to change to TPS3435AFACADDFRQ1 since the current one is out of stock :C

3V3_MAIN
C38 100nF
GND

WDI
GND



If Watchdog Timeout, restarting WDI ticks unlatches WDO

LOW on fault

WDO Low On Fault (Open Drain)

R58 0

3V3_MAIN
R59
GND

3V3_MAIN
U6

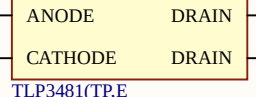
TP1

FAULT 3V3

SN74LVC1G08DCKR

R60 680

K5



TPS35AA17AGADDFR	WD + SVS	A	Adjustable	0	1, 128	Adjustable	open-drain	1.85V
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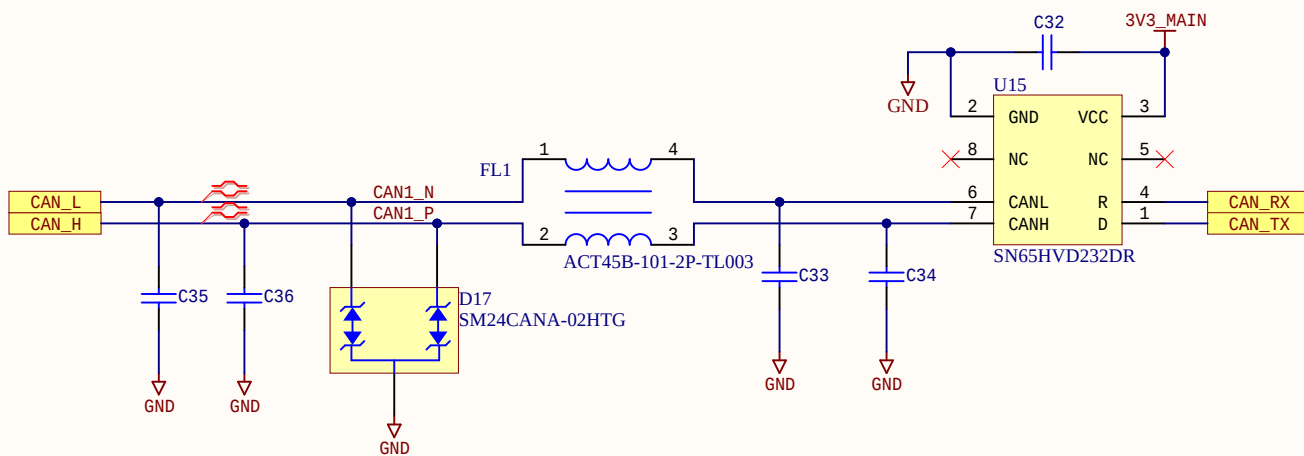
$$t_D (\text{sec}) = 4.95 \times 10^6 \times C_{CRST} (F) \quad (3)$$

To minimize the difference between the calculated reset delay time and the actual reset delay time, use a high-quality ceramic dielectric COG capacitor and minimize parasitic board capacitance around this pin. [Table](#)

A

B

C



A

B

C

D

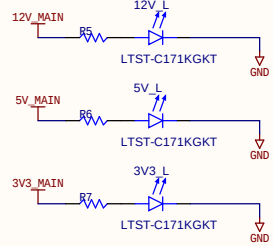
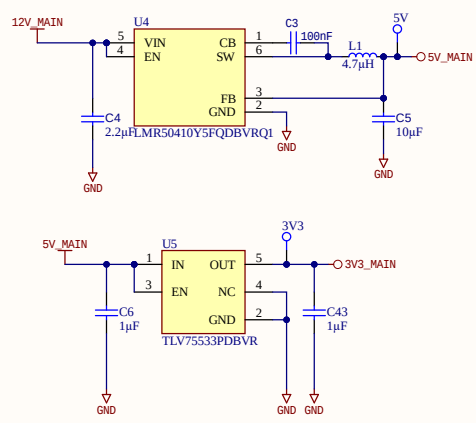
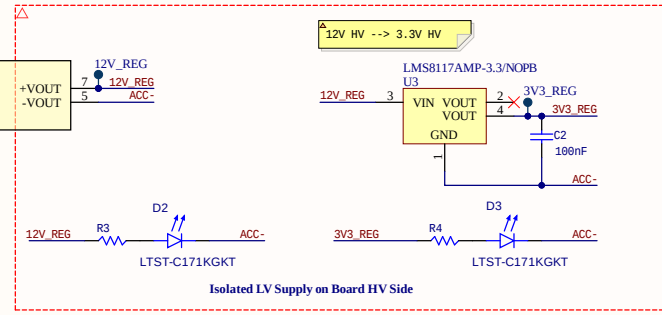
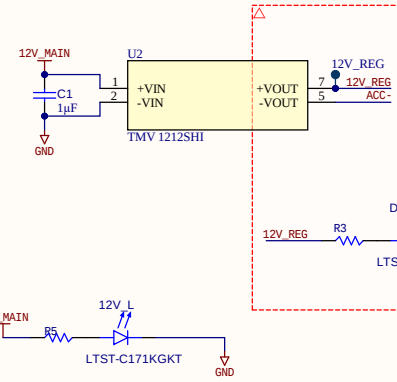
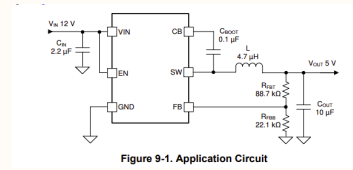
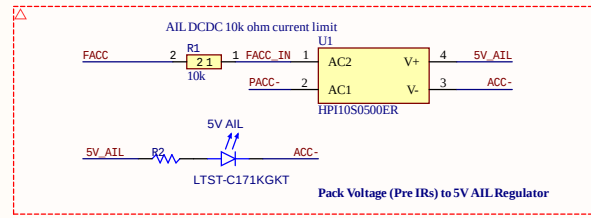
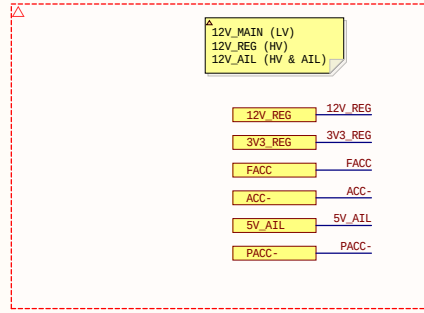
1

2

3

4

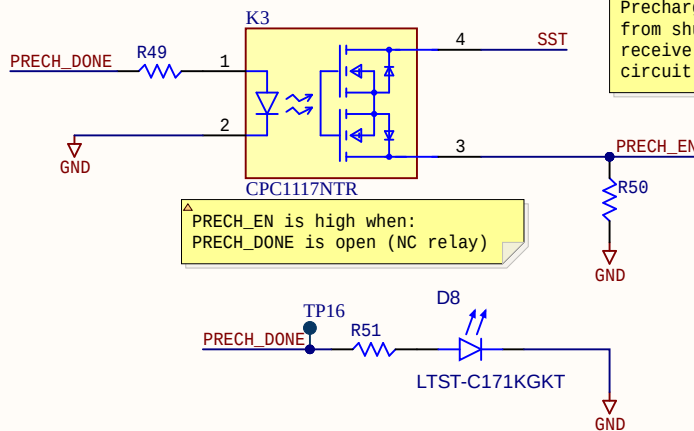
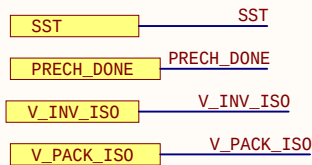
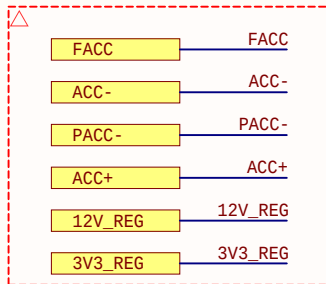
5



A

B

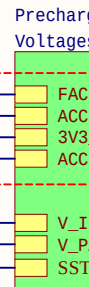
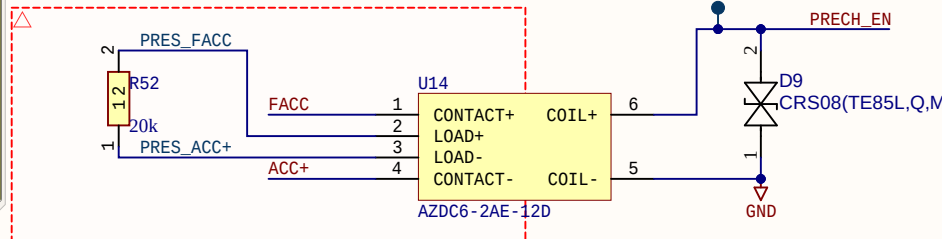
C



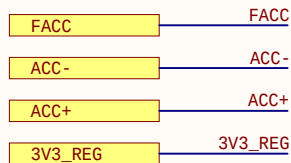
Ensures rules compliance: Precharge relay coil is powered from shutdown circuit (can only receive power when shutdown circuit is present)

This resistor footprint includes mounting hols for heat sink. Other inrush limiting resistor, for AIL DCDC is mounted to the same heatsink on its other side but that component footprint lacks the mounting holes for the heatsink since they both share the same one

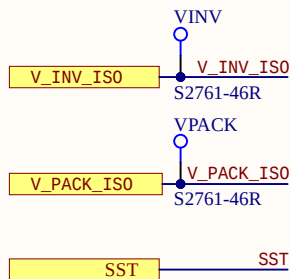
Heat Sink PN: 657-10ABPE (to-247 package)



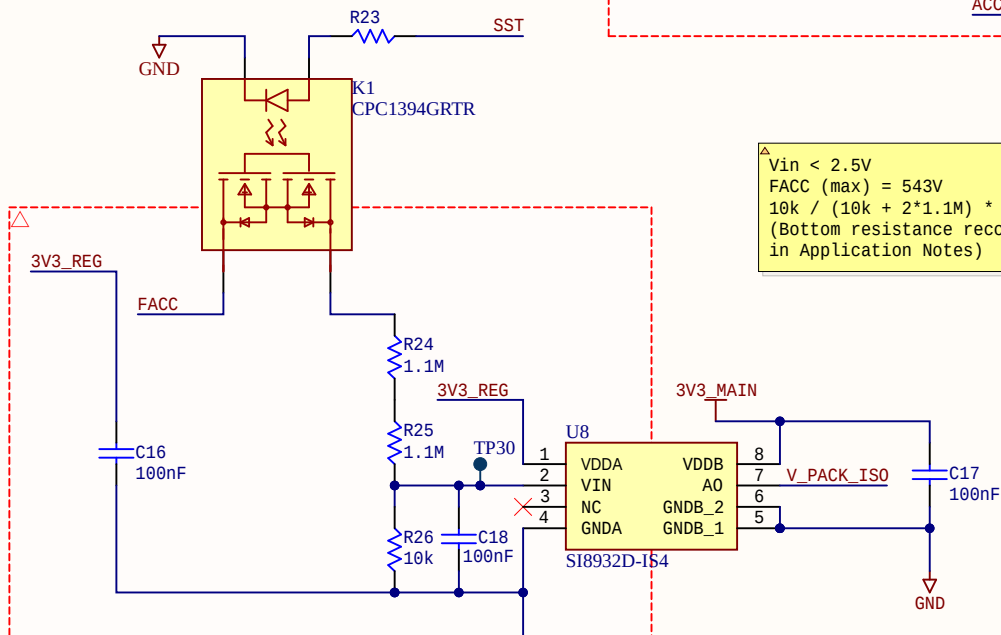
A



B

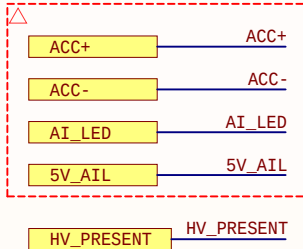


C



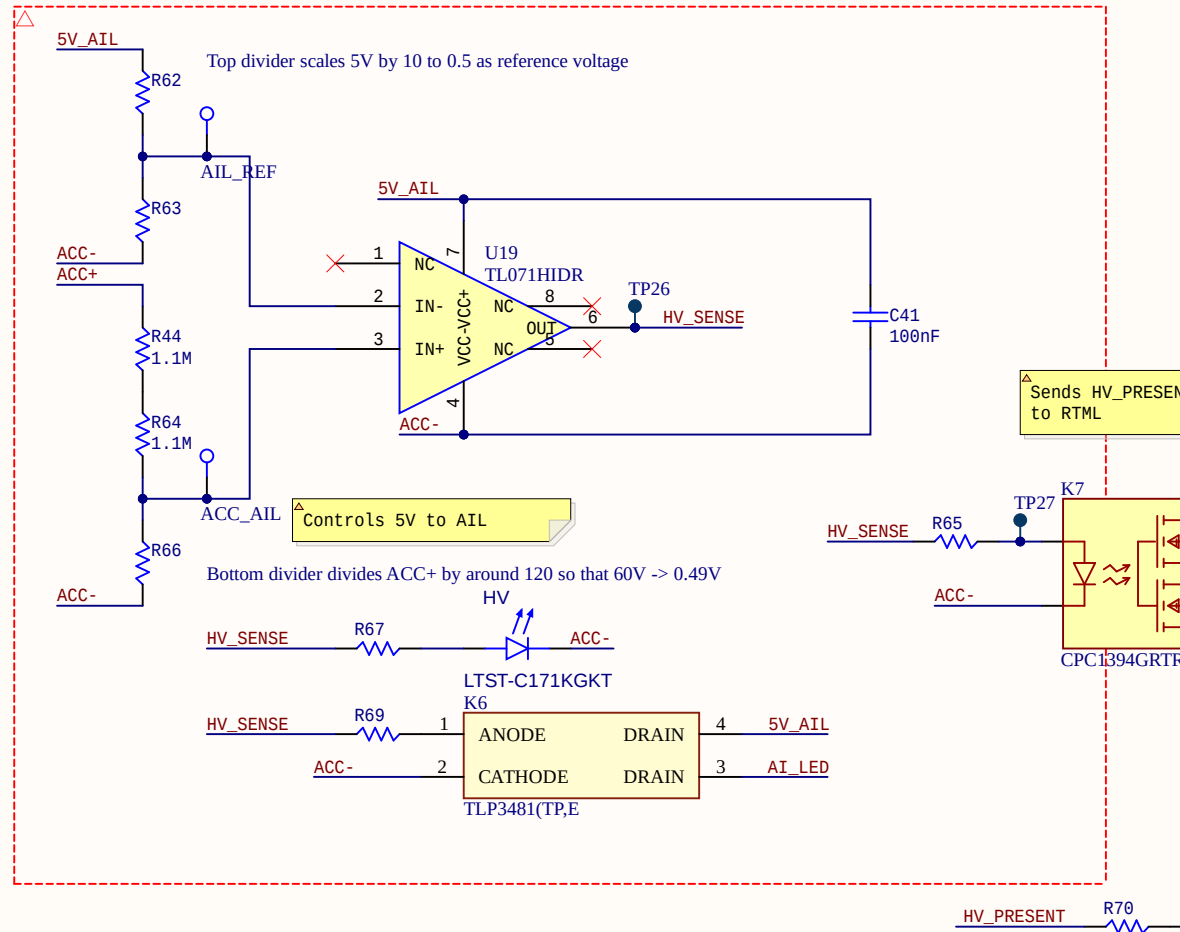
A

Provides HV presence signal (HV_PRESENT) to other boards/AIL



B

C



A

B

C

D

